

Taxi Revenue Maximization – Project Report

Objective

The goal of this analysis was to identify customer payment behaviour and determine whether payment method (Card vs Cash) influences taxi revenue. The findings can help taxi operators encourage higher-revenue payment channels

Key Findings

- Credit card payments are used more frequently than cash payments.
- Customers who pay by credit card generally generate higher average fares compared to cash-paying customers.
- Longer trips and higher fare amounts are more commonly associated with card payments.
- The statistical analysis confirmed that the difference in average fare between card and cash customers is significant and not due to random variation.

Recommendations

To maximize revenue, taxi companies should encourage card payments by:

1. Offering loyalty rewards for card users.
2. Providing discounts through digital payment partners.
3. Making cashless payment options more visible and convenient.
4. Running promotions targeted frequent card users.

Hypothesis Testing

Null Hypothesis (H_0): There is no difference in average fare between card-paying and cash-paying customers.

Alternative Hypothesis (H_1): There is a difference in average fare between card-paying and cash-paying customers.

Test Used - Independent Two-Sample T-Test

Results

- **T-statistics:** 182.30
- **P-value:** 0.0

Since: $p\text{-value} < 0.05$. The null hypothesis is rejected.

There is a **statistically significant difference** between the average fares of card-paying and cash-paying customers.

Conclusion

The analysis demonstrates that **card-paying customers generate significantly higher revenue than cash-paying customers**. Statistical testing confirms that the difference is not due to random chance. Therefore, encouraging digital/card payments can be an effective strategy for **maximizing taxi revenue**.